بسم الله الرحمن الرحيم

السلام عليكم ورحمة الله وبركاته



objectives

- Definition
- Classification
- Kwashiorkor
- Marasmus
- Treatment
- Prognosis





What is Nutrition



Nutrition:

- the provision of all essential nutrients in adequate amounts to insure proper growth and maintenance of body functions
- involves various chemical reactions and physiological transformations which convert foods into body tissues and activities
- involves ingestion, digestion and absorption of various nutrients
- transport into cells
- removal of unusable elements and waste products of metabolism

Malnutrition

Malnutrition:

- Is a pathological state resulting from a relative or absolute **deficiency, or excess** of one or more essential nutrients.
- It is one of the leading causes of morbidity and mortality in children under 5 years of age.

A healthy diet provides a balanced nutrients that satisfy the metabolic needs of the body without excess or shortage

Dietary requirements of children vary according to age, sex and development

Macronutrients (carbohydrates, lipids, proteins) are needed for energy and cell multiplication & repair.

Micronutrients are trace elements & vitamins & minerals, which are essential for metabolic processes.

- Malnutrition include both ends of the nutrition spectrum

under nutrition



overweight



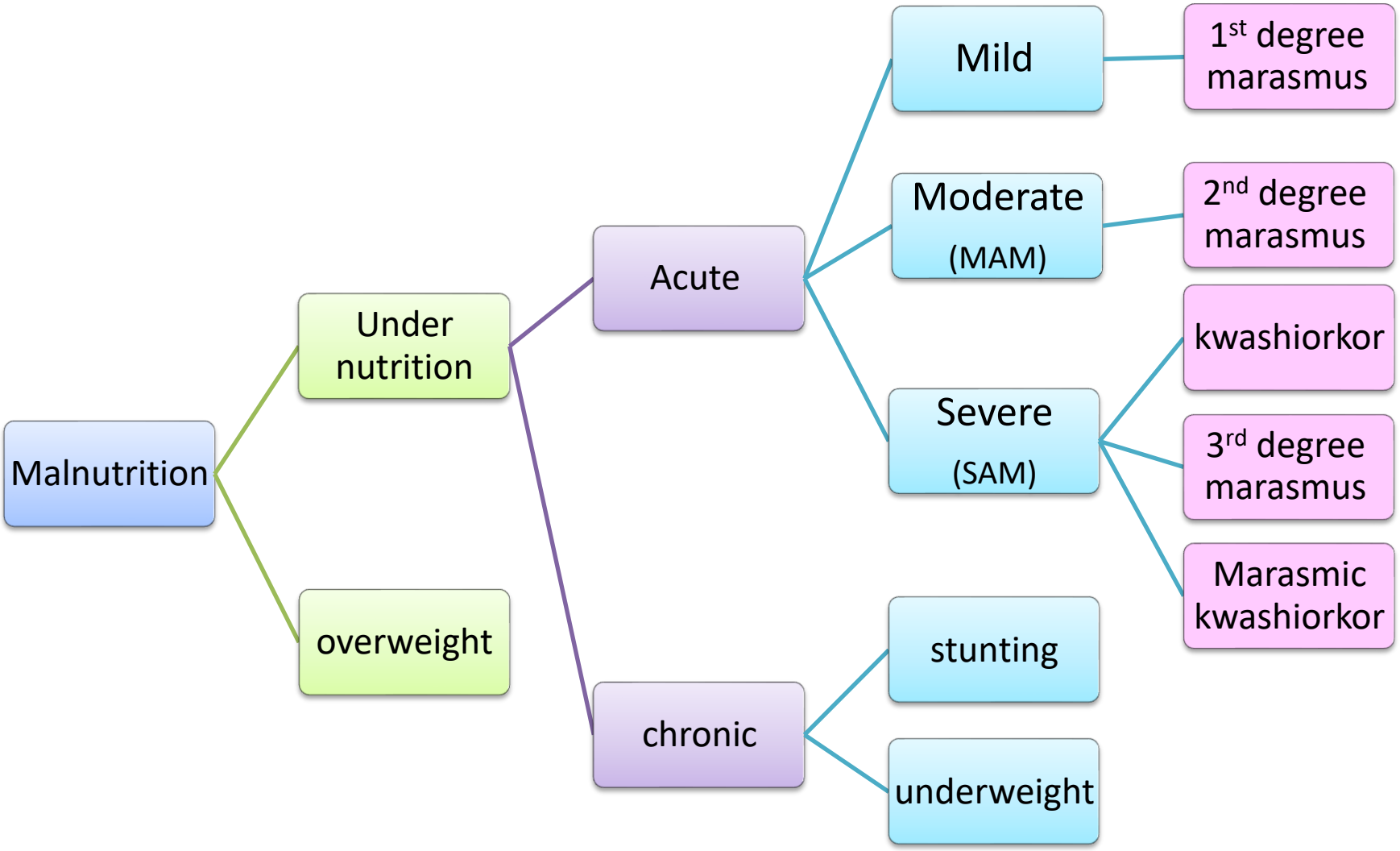
Assessment of nutritional status

- **Feeding history:** type, amount, frequency of food in weaning period and history of repeated starvation during diarrhea.

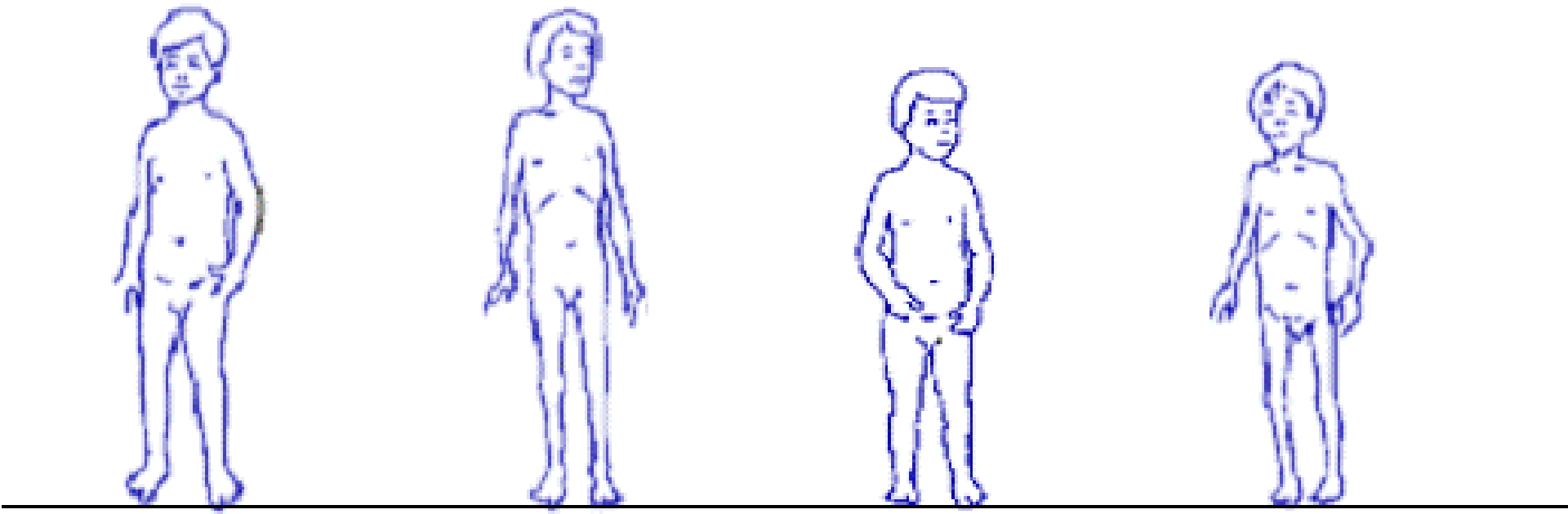
- **Clinical examination** for signs and symptoms of PEM.(protien energy malnutrition).

- **Anthropometric measurements:**
  1. Weight for age
  2. Height for age
  3. Weight for height
  4. Mid-arm circumference
  5. Skin fold thickness
- **Biochemical tests:**
  1. Reduced serum albumin.
  2. Decreased serum transferrin.
  3. Increased non essential /essential amino acid ratio.

Categories of Malnutrition



Macronutrient & Micronutrient Deficiency



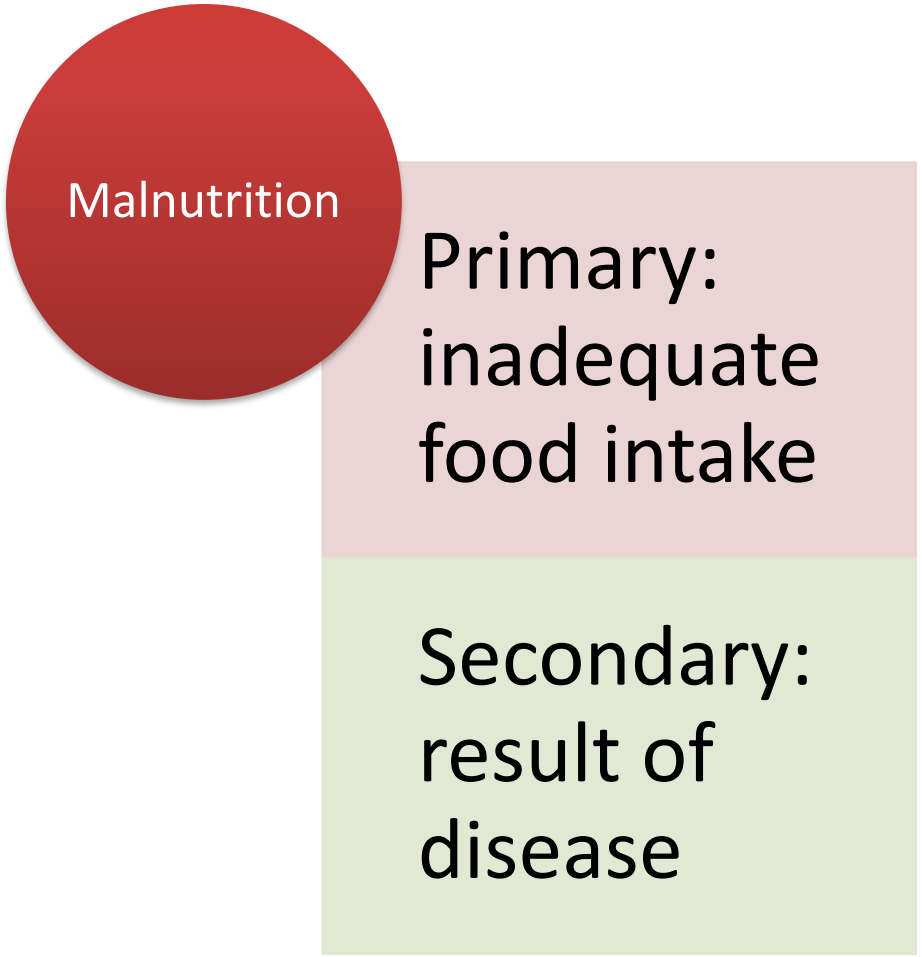
Normal
Normal weight and height

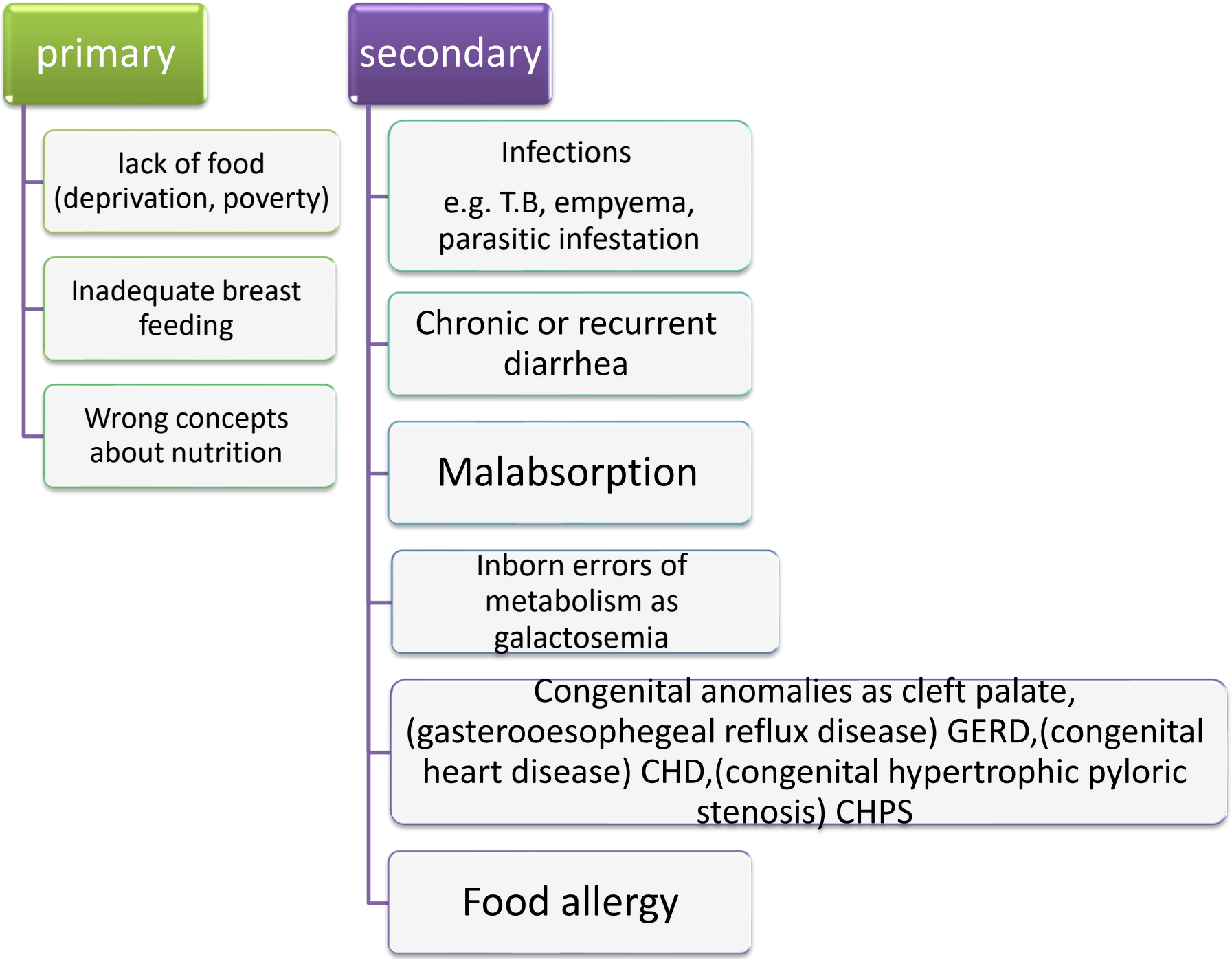
Wasted
Thinner than normal

Stunted
Shorter than normal

Wasted and stunted
Thinner and shorter than normal

Causes of malnutrition




primary
lack of food (deprivation, poverty)
Inadequate breast feeding
Wrong concepts about nutrition
secondary
Infections
e.g. T.B, empyema, parasitic infestation
Chronic or recurrent diarrhea
Malabsorption
Inborn errors of metabolism as galactosemia
Congenital anomalies as cleft palate, (gasterooesophegeal reflux disease) GERD,(congenital heart disease) CHD,(congenital hypertrophic pyloric stenosis) CHPS
Food allergy

MILD MALNUTRITION

Mild Malnutrition : commonest form especially in post weaning period (6m-2y).

Main features:

1. Growth failure:

   A- slowing , cessation or loss of weight.

. B- slowing, cessation in linear growth. normal or low weight /height ratio.

   C- Decreased in mid arm circumference.

   D- Normal or decreased skin fold thickness.

   E- Retarded bone maturation.

- 2- High rate of infection.
- 3- Decreased activity .
- 4- Retardation of motor development
- 5- Anemia

KWASHIORKOR

- **Definition:**
- Term kwash. taken from language of Ghana refers to deposed child .It results mainly from lack of protein
- **Etiology :**
- Dietetic error : feeding of child with an imbalanced diet very low in protein.
- Infections :diarrhea , respiratory & parasitic infestations play an important role in onset of kwash , in already malnourished children.

Clinical picture :

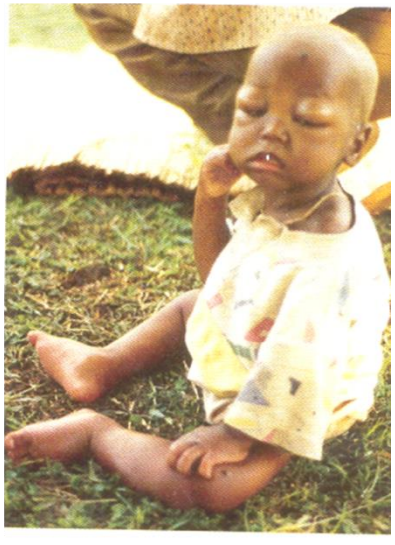
- 6 m. - 3 y.
- Doll facies
- Apathy
- Growth failure
- Muscle wasting
- Edema
- Cracky paint dermatosis
- Flag sign



Anorexia
Diarrhea
Abdominal distension
Hepatomegaly
Anemia
Vitamin A , B deficiency





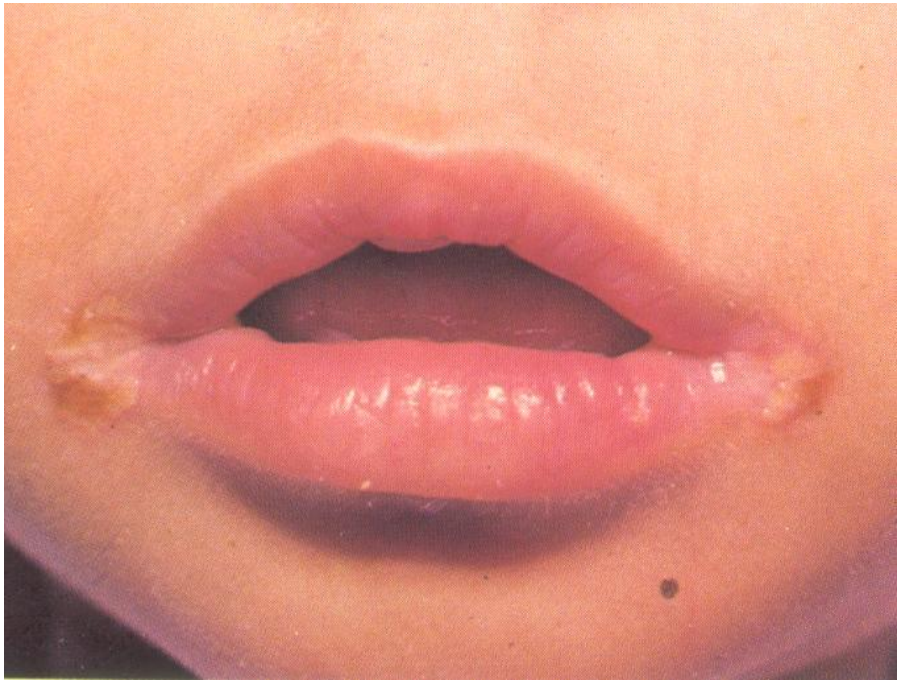








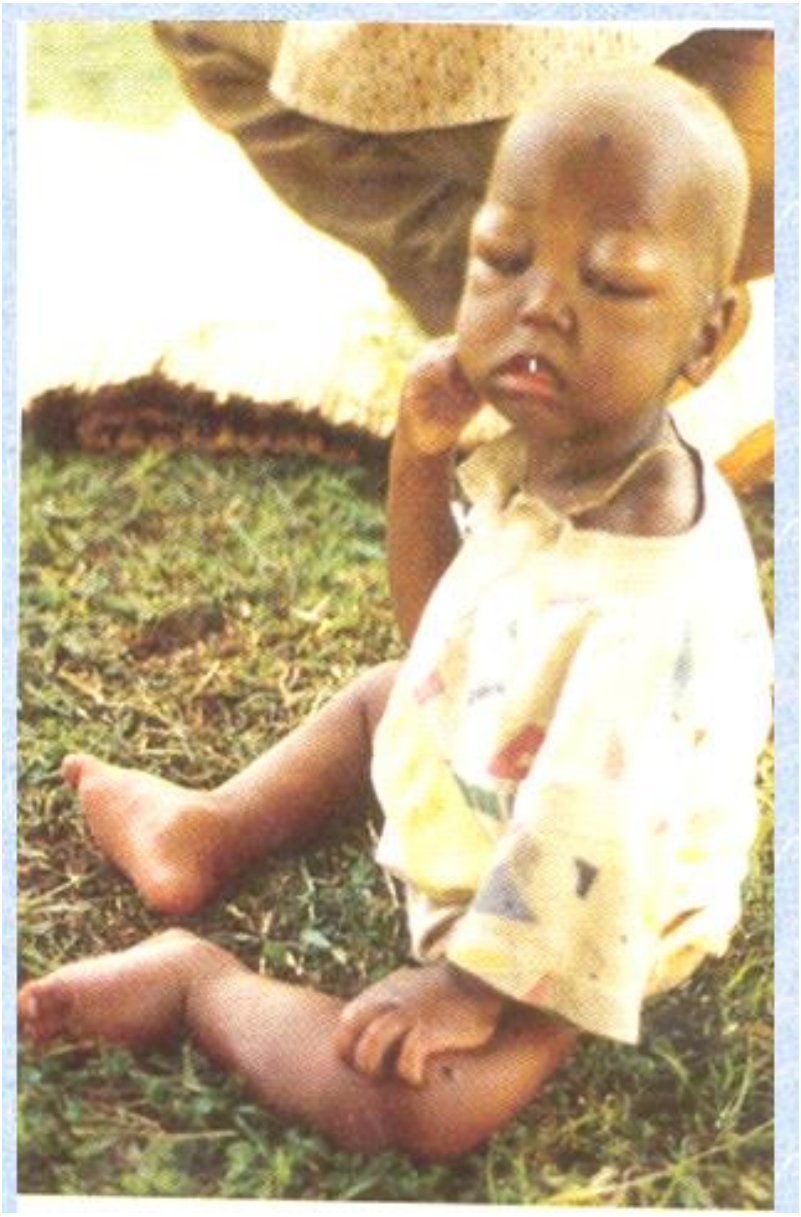












Constant feature	Pathognomonic	Common signs
• Apathy • Edema • Anorexia • Growth failure • Hypoproteine-mia • Fatty liver	• Apathy • Falg sign • Carcky paint dermatosis	• Anemia • vit deficien • Skin change • Hair change • infection

- **Differential diagnosis of kwash:**

1- Edema of kwash,differ.from other causes of edema (renal ,cardic and hepatic)

2- Differential diagnosis of dermatitis (skin lesions)

- ❖ Napkin dermatitis .
- ❖ Monilial infection
- ❖ Acrodermatitis enteropathica.

Laboratory manifestations

- Decreased level of

✓ Albumin

✓ Amino acids

✓ Cholesterol , triglycerides

✓ Na , k , Mg , Cu

✓ Amylase , lipase , ALK Ph

Marasmus

- DEF: It is a chronic state of under nutrition with progressive loss of weight, gross wasting of muscles &subcutaneous fat.
- **Etiology :**
- Dietetic (nutritional marasmus)
- Non dietetic causes of marasmus.



Clinical manifestation :

- From birth-3 **y.**

Clinical Features of Marasmus

- Severe wasting of muscle & s/c fats
- Severe growth retardation
- Child looks older than his age
- No edema or hair changes
- Alert but miserable
- Hungry
- Diarrhea or constipation





Characteristics of Marasmus and Kwashiorkor

Marasmus



- Severe weight loss and wasting
- Ribs prominent
- Limbs emaciated
- Muscle wasting
- May have good appetite
- With correct treatment, good prognosis

Kwashiorkor



- Bi-lateral oedema and fluid accumulation
- Loss of appetite
- Brittle thinning hair
- Hair colour change
- Apathetic and irritable
- Face may seem swollen
- High risk of death

Biologic differences

Marasmus

- ✓ Weight loss
- ✓ Normal/low extracellular water
- ✓ Normal or low protein
- ✓ Normal enzymes
- ✓ Normal transaminases

Kwashiorkor

- ✓ No weight loss
- ✓ High extracellular water
- ✓ Low protein
- ✓ Low enzymes
- ✓ High transaminases

Marasmic kwashiorkor

- Combined form severe malnurition .
- Starts as marasmus with loss of weight more than 40% of standard , suddenly put on a starchy diet vey low in protein signs of kwash develop.

Investigations for PEM

- Full blood counts
- Blood glucose profile
- Septic screening
- Stool & urine for parasites & germs
- Electrolytes, Ca, Ph & ALP, serum proteins
- CXR & tuberculin test
- Exclude HIV & malabsorption

Complication of under nutrition

1) Death
2) Impair immune system
3) Increase risk of infections
4) Hypoglycemia
5) Hypothermia
6) Hypokalemia
7) Hyponatremia
8) Heart failure
9) Dehydration & shock

Other non fatal complications

a- Permanent affection of cognition
b- Short stature
C- Blindness due to vitamin A deficiency.

TREATMENT



A. GENERAL PRINCIPLES FOR ROUTINE CARE (the '10 Steps')

There are ten essential steps:

1. Treat/prevent hypoglycemia
2. Treat/prevent hypothermia
3. Treat/prevent dehydration
4. Correct electrolyte imbalance
5. Treat/prevent infection
6. Correct micronutrient deficiencies
7. Start cautious feeding
8. Achieve catch-up growth
9. Provide sensory stimulation and emotional support
10. Prepare for follow-up after recovery

TTT of hypoglycemia

If the child is conscious and dextrostix shows <3mmol/l or 54mg/dl			
50 ml bolus of 10% glucose or 10% sucrose solution (orally or by nasogastric tube)	Then feed starter F-75 every 30 min for two hours.	Then two-hourly feeds, day and night	give antibiotics from the start

Treatment of hypothermia

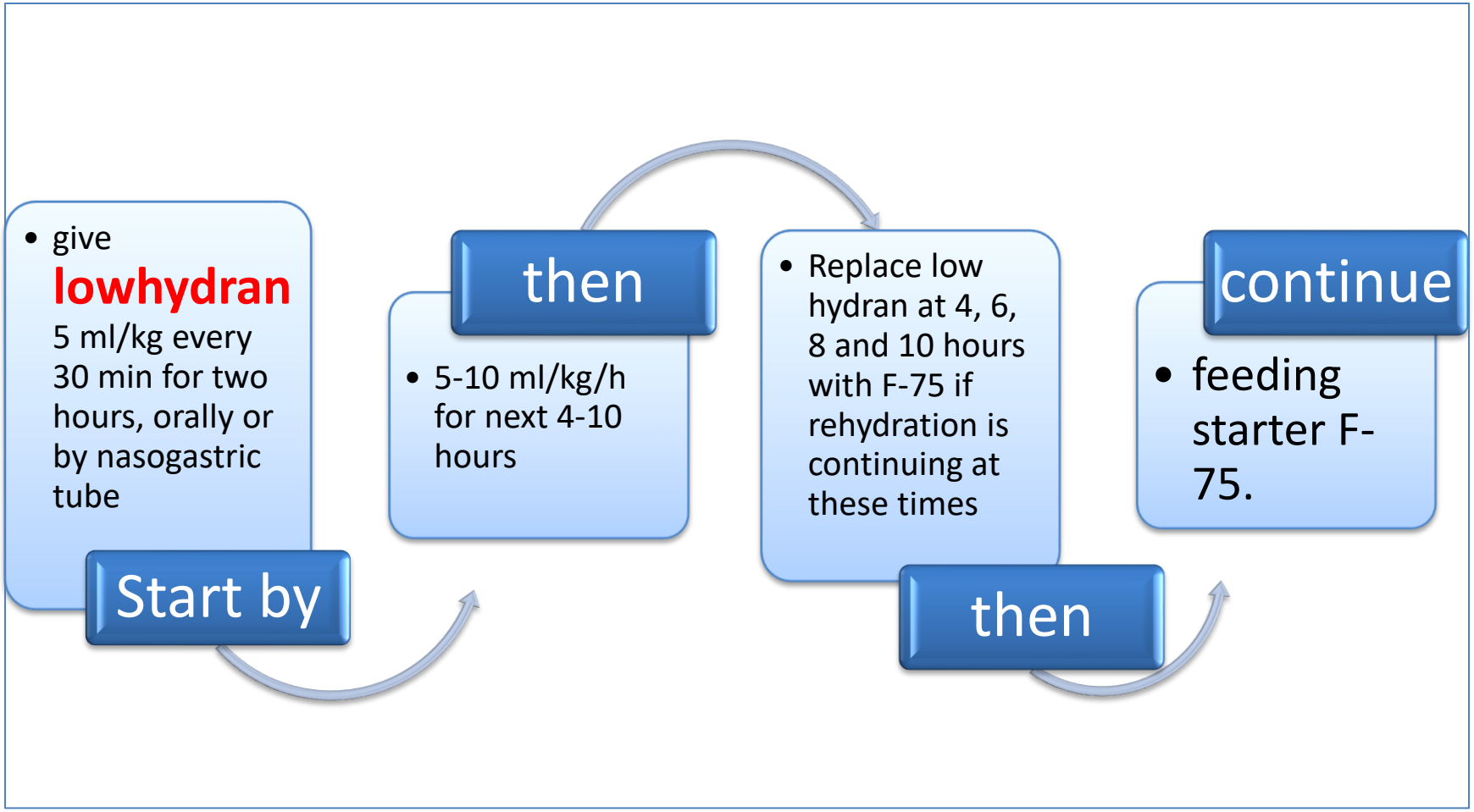
Feed straight away (or start rehydration if needed)

Rewarm the child



Give antibiotics

TTT of dehydration



TTT of electrolyte imbalance

- so give:
- extra potassium 3-4 mmol/kg/d.
- extra magnesium 0.4-0.6 mmol/kg/d.
- when rehydrating give low sodium rehydration fluid (e.g. low hydran).
- prepare food without salt.

5. Treat/prevent infection

- Give routinely on admission broad-spectrum antibiotic(s) and measles vaccine if child is >6m and not immunized (delay if the child is in shock).



6. Correct micronutrient deficiencies:

- All severely malnourished children have vitamin and mineral deficiencies.

- Give:
  - Vitamin A :orally on Day1
    - for age >12 months give 200,000 IU
      for age 6-12 months give 100,000 IU
      for age 0-5 months give 50,000 IU





- Give daily for at least 2 weeks:
  - Multivitamin supplement
  - Folic acid 1 mg/d (give 5 mg on Day 1)
  - Zinc 2 mg/kg/d
  - Copper 0.3 mg/kg/d
  - Iron 3 mg/kg/d but only when gaining weight



7. Start cautious feeding

- Why cautious feeding in the stabilization phase?

  Because of

  - the child's fragile physiological state
  - reduced homeostatic capacity.

The essential features of feeding in the stabilization phase are:

Small, frequent feeds of low osmolarity and low lactose oral or nasogastric (NG) feeds (never parenteral feeding)

100 kcal/kg/d

1-1.5 g protein/kg/d

130 ml/kg/d of fluid

if the child is breastfed, encourage to continue breast feeding but give the prescribed amounts of starter formula to make sure the child's needs are met.

8. Achieve catch-up growth

- In the rehabilitation phase a vigorous approach to feeding is required to achieve very high intakes and rapid weight gain of >10 gm gain/kg/d.

Severe anemia in malnourished children

- blood transfusion is required if Hb is less than 4 g/dl

  Or

- if there is respiratory distress and Hb is between 4 and 6 g/dl.

- Give:
  - whole blood 10 ml/kg body weight slowly over 3 hours
  - furosemide 1 mg/kg IV at the start of the transfusion

Indications of discharge

- Children with severe acute malnutrition should only be discharged from treatment when their:
  - ❑ weight-for-height/length is ≥–2 Z-score and they have had no edema for at least 2 weeks.

  Or

  - ❑ mid-upper-arm circumference is ≥125 mm and they have had no edema

PROGNOSIS

- Kwashiorkor & Marasmus-Kwashiorkor have greater risk of morbidity & mortality compared to Marasmus and under weight
- Early detection & adequate treatment are associated with good outcome
- Late ill-effects on IQ, behavior & cognitive functions are doubtful and not proven


Thank
You
Sherifa